

Chapter-1: Safety-Oriented Traffic Conflict Detection and Resolution for Autonomous Vehicles at Unsignalised Intersections Using CARLA Simulation

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Abstract

Unsignalised intersections are still challenging for AVs, as right-of-way cannot be determined through static signal phase but require inference from dynamic interaction among various parties. In this paper, a safety-centric approach is introduced which incorporates time-to-collision (TTC) and distance-at-risk (DAR) together to detect both immediate conflicts and potential conflicts. An hierarchical controller classifies the interactions to safe state, potential-conflict state and imminent-conflict state, respectively, and generates a corresponding manoeuvre such as maintaining normal trajectory, adjusting speed and trajectory in a bounded manner and yielding to other parties. The proposed approach is evaluated based on 600 episode-pairs using CARLA 0.9.14 under low, medium and high traffic density in situations including crossing, merging, turning and continuous arrival. Under the assumptions of the simulation, the fusion detector achieves a classification accuracy of 98.7% and conflict resolution rate of 99.2%, while the rule-based baseline reaches 85.6% and 81.0%, respectively. Meanwhile, the mean TTC after the resolution increases to 6.8 s whereas the mean time spent at the intersection remains to be 4.9 s. The sensitivity analysis shows that the performance deteriorates gradually under observation noise.

Keywords: Autonomous Vehicles; CARLA; Conflict Detection; Unsignalised Intersection; Safety-Oriented Control.

1. Introduction

Autonomous driving needs to deal with the case of multiple vehicles that can potentially occupy the same road space at a given instant. In an un-signalized intersection, the lack of a traffic phase implies that perception, motion prediction, interaction analysis, and control become key. The situation is even more challenging when all the three variables, namely speed, intention, and order of arrival of the vehicles vary simultaneously. Planning is always discussed as a tightly-coupled process in which the uncertainty from perception and motion prediction has to be fed into the decision-making process (Lefèvre et al., 2014; Paden et al., 2016; Pendleton et al., 2017; Schwarting et al., 2018).

The technique of conflict analysis makes it possible to analyze traffic safety without accidents' occurrence. Surrogate measures of traffic safety include Time-to-Collision (TTC) and Post-Encroachment Time (PET), as they explain the collision process and are widely applied in microscopic traffic safety analysis (Gettman & Head, 2003; Laureshyn et al., 2010; Mahmud et al., 2017; Wang et al., 2021). Though the TTC concept works well in a situation where predicted paths intersect at some point, corresponding to collision time, generally speaking, TTC may either be indeterminate or overestimate danger when curves, converging or nearly parallel paths are under consideration. Additional metric in the space domain will increase clarity of dangerous clearance decrease recognition.

This paper proposes a TTC-DAR hierarchical method for the problem of collision avoidance and mitigation. The contributions of this research can be summarized as follows. Firstly, the integration of temporal and spatial safety metrics makes it possible to represent both imminent and emergent collisions. Secondly, the lexicographic safety criterion is employed; namely, the proposed control strategies have to

satisfy the requirements of clearance, lane, acceleration and jerk safety criteria prior to those of time-delay and energy efficiency. Thirdly, all controllers are tested in the same CARLA settings.

2. Related Work and Theoretical Basis

2.1 Surrogate safety and conflict indicators

Traffic conflict theory is defined as any situation involving encounters whereby the road users are in proximity with one another and therefore there is a need for some kind of avoidance maneuvers. According to previous literature, TTC was thought to be a scale for the measurement of hazard, while modern research has focused more on the temporal measure of traffic conflicts as well as simulation applications in traffic conflict to crashes (Mahmud et al., 2017; Wang et al., 2021; Gettman & Head, 2003). The benefit of such measures is that they can be measured more frequently than crashes.

The suggested measure of DAR is the minimum predicted clearance between boundaries of two oriented vehicle envelopes over a period of three seconds. This measure is not meant to replace TTC. Instead, it detects interactions where future separation will be low even in situations where TTC is not defined. Both measures should be used based on the more general rule that safety analysis needs to be based on complementary information with explicit assumptions about domain and thresholds (Wang et al., 2021).

2.2 Decision-making at intersections

Methods for intersection management are varied and include priority rules, reservation and coordination schemes, reachability analysis, optimization, and learning techniques. The main advantage of the rule-based approach is low computational complexity, while the main disadvantage is brittleness if the order of arrivals is unclear or if several vehicles give way at once. Methods of reservations and coordination have higher throughput if communication works reliably (Dresner & Stone, 2008; Rios-Torres & Malikopoulos, 2017). Reachability and formal safety techniques guarantee better safety properties, but they might make conservative assumptions or incur higher computational costs (Althoff & Dolan, 2014; Shalev-Shwartz et al., 2017).

Table 1. Comparison of intersection-conflict management approaches

Approach	Main advantage	Principal limitation
Rule-based priority	Fast, interpretable, easy to audit	Brittle in dense or ambiguous interactions
Single surrogate measure	Low computational burden	Can miss latent spatial conflicts
Reservation/coordination	Efficient when communication is reliable	Depends on connectivity and protocol compliance
Learning-based policy	Adaptive interaction behavior	Safety may degrade outside training data
Proposed TTC–DAR hybrid	Graded and auditable intervention	Requires threshold calibration and validation

3. Proposed Safety-Oriented Framework

3.1 Architecture and state processing

The controller is embedded in a synchronous receding-horizon cycle. At each time step of 0.05 s, the controller gets the pose, velocity, acceleration, heading, and geometry of lanes of both the ego and the surrounding vehicles. Using a reachability gate, actors not entering the conflict area in the prediction horizon are filtered out. The remaining actors are then propagated using constant velocity or constant

acceleration models depending on recent acceleration stability. Vehicle envelopes are sampled for three seconds in an oriented manner, and then TTC and DAR are calculated.

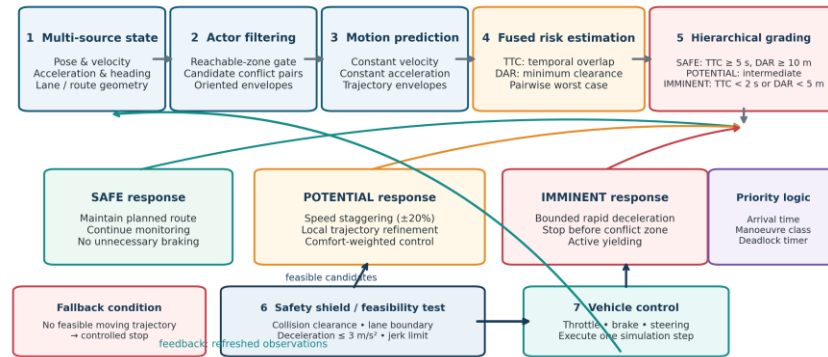


Figure 1. Safety-oriented TTC–DAR detection and resolution architecture

The processing flow from state acquisition to execution control is shown in Figure 1. The design decouples the prediction, risk assessment, response selection, and protection phases. The decoupling ensures that the triggering mechanism for every intervention can be traced, and enables the controller to resort to a controlled stop when there is no moving path satisfying the hard constraints.

3.2 Risk calculation and grading

TTC for two vehicles i and j is the earliest time in the future when the envelopes will collide. In case there is no envelope collision during the horizon, the TTC is considered to be infinite. The DAR is defined as the minimum clearance of the envelope over the horizon. Imminent conflicts are declared in case TTC is less than 2 seconds or DAR is less than 5 meters. Potential conflicts are declared in case TTC is more than 2 seconds but less than 5 seconds or DAR is more than 5 meters but less than 10 meters. The state is said to be safe only in case both these metrics stay above their respective safe limits.

Table 2. Hierarchical conflict grading and prescribed response

Risk state	TTC	DAR	Controller response
Imminent	< 2 s	< 5 m (either indicator sufficient)	Bounded rapid deceleration and active yield
Potential	$2 - < 5$ s	$5 - < 10$ m	Speed staggering and local path refinement
Safe	≥ 5 s	≥ 10 m	Maintain route and continue monitoring

3.3 Resolution policy

The selection of actions is based on lexicography. At the first level, all the actions that lead to violation of the clearance of collision, the lane, the acceleration limit of 3m/s^2 , and the jerk constraints are rejected. Then, for feasible actions, the controller chooses the one that maximizes the minimum estimated TTC and DAR. Optimization of delay and control input is done only when safety requirements have been achieved. Conflicts can be normally handled by changing speeds of the vehicle at about 20% of its current speed so as to avoid conflicts through separation of arrival times. Immediate conflicts are handled using bounded braking and yielding.

The deterministic priority rule will ensure that mutual-yield deadlock does not occur during simulation. Linear maneuvers have priority over conflicting turns and early arrival breaks any ties. The waiting timer

provides a last-resort deterministic tie-breaker. This mechanism allows for reproducible comparison, but it is not proposed as a viable communication protocol for the road; rather, explicit negotiation is recommended.

4. CARLA Experimental Method

4.1 Environment and scenarios

The experiments have been carried out using CARLA 0.9.14 in the synchronous mode on the map Town04. Fixed step size is equal to 0.05 sec, prediction time horizon is 3.0 sec, while each episode lasts until all vehicles leave the area of interest or until 120 seconds pass. Routes, spawn transforms, weather parameters, vehicle blueprints, and random seeds are saved in episode metadata. Warm-up phase guarantees the stabilization of traffic actors before entering the area of interest. The controllers get the same state vector based on ground-truth data before any noise injection.

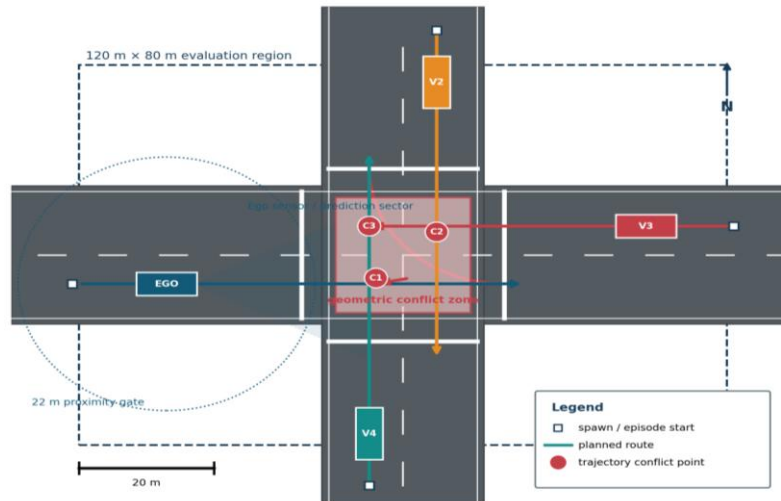


Figure 2. CARLA unsignalised four-way intersection test configuration

The testing dataset comprised 600 pairs of events, namely 252 crossing events, 186 merging events, 102 turning-interaction events, and 60 diverging or control events. The controllers were evaluated against the same set of initial conditions. Traffic density was categorized into low, medium, and high levels, while approach speed, arrival offset, route choice, and behavior seed were varied. The offline labels were computed using future oriented-envelope intersection and clearance criteria at the native simulation frequency. The labels were not available to the online controllers.

Table 3. Main simulation and evaluation settings

Parameter	Setting
Simulator and map	CARLA 0.9.14, Town04
Execution	Synchronous, 0.05 s step
Prediction horizon	3.0 s
Episodes	600 paired episodes
Traffic density	Low, medium, high
Baselines	Rule-based and single-TTC controllers
Noise tests	0%, 5%, 10%, and 15% state perturbation

4.2 Metrics and analysis

Accuracy of detection was defined as the percentage of correctly classified conflict/non-conflict episodes. For resolution to be successful, there had to be successful completion of the maneuver without collision

and TTC/DAR levels had to be brought back to above safe values before the ego vehicle left the evaluation area. The passage time was computed from the time the ego vehicle reached the approach boundary to when it cleared the intersection. Post-resolution mean TTC and maximum deceleration gave information on safety and comfort. All the controllers had the same starting conditions and hence comparison was done at the episode level.

5. Results and Discussion

5.1 Detection and resolution performance

The detection rate of the fused sensor was 98.7%, compared with 89.2% for the single-TTC technique and 85.6% for the rule-based baseline. The 9.5% increase from TTC only implies that the spatial clearance provided information beyond immediate temporal proximity. Error analysis confirmed this: some instances of turning and merging maneuvers passed near a point of future intersection without a TTC yet, where DAR detected danger first.

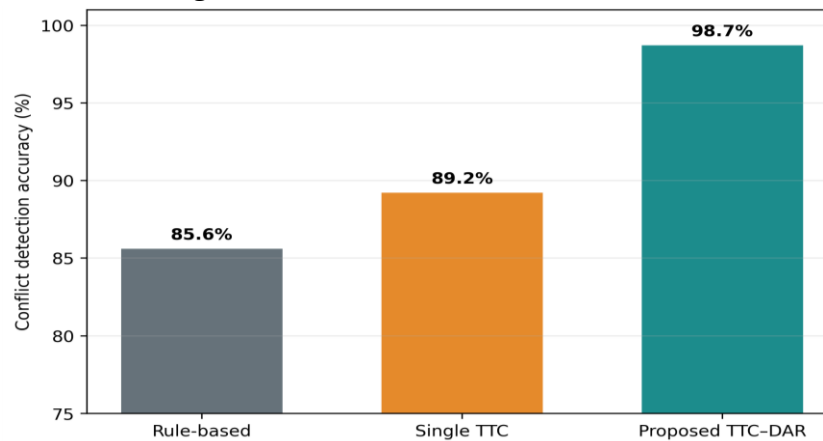


Figure 3. Conflict-detection accuracy across decision methods

Resolution rate was achieved at 99.2%, while single TTC and rule-based controllers had rates of 88.0% and 81.0% respectively. The mean post-resolution TTC was also raised to 6.8 seconds, and the number of dangerous conflicts reduced to 7.7% of the number in rule-based controls. The proposed controller took an average of 4.9 seconds to complete the intersection process: 0.7 second longer than the aggressive rule-based system, but 0.9 second quicker than the conservative TTC-only controller.

Table 4. Aggregate performance across 600 paired episodes

Metric	Rule-based	Single TTC	Proposed TTC-DAR
Detection accuracy (%)	85.6	89.2	98.7
Mean post-resolution TTC (s)	3.2	4.1	6.8
Resolution success (%)	81.0	88.0	99.2
Residual dangerous conflicts (relative)	100	61	7.7
Mean passage time (s)	4.2	5.8	4.9

5.2 Robustness, ablation, and limitations

The effect on accuracy degraded steadily as noise and traffic density rose. The performance varied from 99.4% for low-density and noiseless cases to 94.7% for high density and 15% perturbation. This is

encouraging as there was no collapse, but even the most difficult case was not enough for unrestricted use. The high-density errors occurred mostly due to the rapid change of acceleration and actors' association, neither of which was well-represented using the deterministic constant velocity/acceleration model.

The absence of DAR caused the performance to drop closer to the TTC-alone baseline, and especially impacted the performance on turning and merging. The false positives occurred mainly at the 10 m boundary, where the trajectory was coming near and then diverging. Although hysteresis decreased repetitive state changes, the threshold tuning remains a practical trade-off; a conservative threshold will allow for early action but will be less efficient, and vice versa.

Results need to be considered within the simulation framework. CARLA does not simulate all uncertainties that exist for sensing, tires, weather conditions, human behaviors, and actuation on the road. Ground truth – based inputs make the comparison an algorithm assessment and not a full-stack perception one. The simulator does not incorporate pedestrians, cyclists, uncertainties due to occlusions, loss of communications, and different sets of legal rules. Lastly, the percentages provided come from the scenario distribution in the experiment and not from a world estimate of crashes prevented.

6. Conclusion

This study presented a safety-oriented conflict-management framework for autonomous vehicles at unsignalised intersections. The method combines TTC with DAR, grades interactions into safe, potential, and imminent states, and applies a proportionate response under hard safety constraints. In 600 paired CARLA episodes, the fused detector improved classification and resolution outcomes relative to rule-based and TTC-only baselines while maintaining competitive passage time. Ablation results indicate that DAR is most useful in geometries where spatial clearance becomes hazardous before a finite collision time is available.

Future work should replace point predictions with probabilistic motion envelopes, calibrate thresholds by maneuver and speed, and integrate explicit connected-vehicle negotiation. Evaluation should expand to vulnerable road users, occlusion, adverse weather, multiple maps, hardware-in-the-loop control, and closed-course testing. Safety claims should ultimately be supported by scenario-based assurance arguments aligned with functional-safety and safety-of-the-intended-functionality standards (International Organization for Standardization, 2018, 2022; SAE International, 2021).

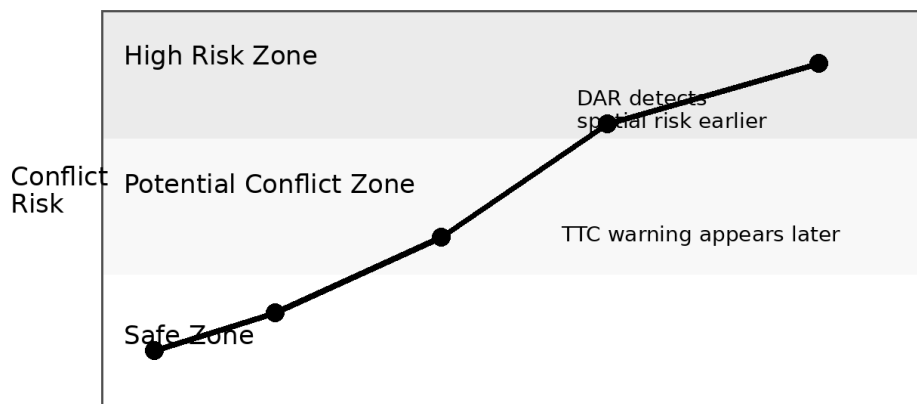


Figure 4. Conceptual evolution of fused TTC–DAR conflict risk

Figure 4 highlights the conceptual connection between temporal and spatial surrogate measures of safety within the suggested TTC-DAR framework. TTC detects situations in which collision time is significant, whereas DAR represents diminishing physical distance prior to the emergence of temporal conflict. The use of both surrogate measures is essential for gradual intervention and enhances the transparency of decision-making of AVs at uncontrolled intersections. Figure 4 is an illustration that supplements the experiment conducted in CARLA by showing why joint risk assessment may lead to better conflict detection.

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